

App Dev Project Report (MAD 1)

1. Student Details

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About Me:

I am a Diploma Level student in the BS Degree Program in Data Science and Applications at Indian Institute of Technology Madras. I am highly interested in technology, data science, and building real-world solutions that solve practical workflow problems.

2. Project Details

Project Title: Placement Portal Application

Problem Statement

To design and develop a role-based web application that digitizes and streamlines the campus placement workflow. The system enables administrators to manage students, companies, and job postings; companies to publish job opportunities and process applications; and students to apply for jobs and track their application status.

Approach

The system was developed using the Flask web framework following a modular MVC-style structure. The application implements authentication, database-driven workflows, approval pipelines, and role-based dashboards to ensure a structured and secure placement management process.

The system integrates SQLite for persistent storage and Jinja2 templating for dynamic frontend rendering.

3. AI / LLM Declaration

AI tools (ChatGPT and Gemini) were used selectively for debugging assistance, conceptual understanding, UI improvement ideas, and documentation formatting.

The overall system design, database modeling, feature implementation, testing, integration, and workflow logic were developed independently.

AI was used as a learning aid to understand issues and implement solutions manually, not for direct copy-paste development.

Extent of AI usage: Approximately 10–15%.

I am comfortable explaining all parts of the codebase during the viva.

4. Technologies and Frameworks Used

Technology / Library	Purpose
Flask	Core backend web framework for routing and request handling
Flask-SQLAlchemy	ORM for database modeling and interactions
SQLite	Lightweight relational database for persistent storage
Jinja2	Template engine for rendering dynamic HTML pages
HTML5 / CSS3	Frontend structure and styling
Bootstrap	Responsive UI design and layout

5. Database Schema Design

Tables

- Admin** — Stores administrator credentials
Fields: id (PK), email (Unique), password

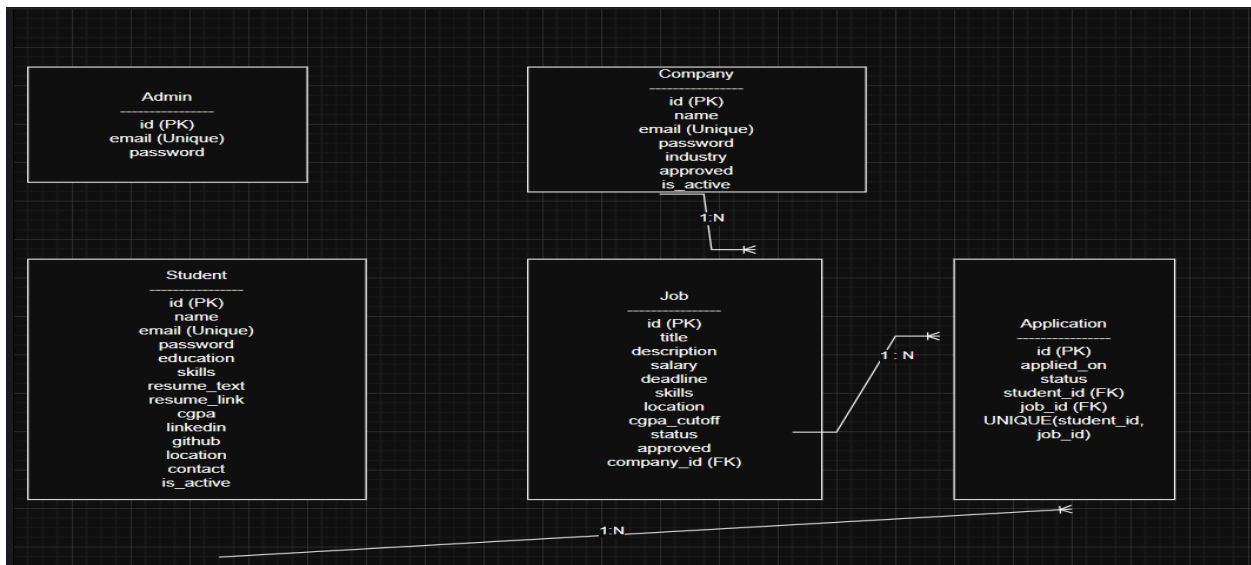
2. **Company** — Stores company profiles
Fields: id (PK), name, email (Unique), password, industry, approved, is_active
3. **Student** — Stores student profiles
Fields: id (PK), name, email (Unique), password, education, skills, resume_text, resume_link, cgpa, linkedin, github, location, contact, is_active
4. **Job** — Stores job/placement postings
Fields: id (PK), title, description, salary, deadline, skills, location, cgpa_cutoff, status, approved, company_id (FK)
5. **Application** — Bridge table for job applications
Fields: id (PK), applied_on, status, student_id (FK), job_id (FK)

Constraints

- Unique email constraint on Admin, Company, and Student
- Foreign key constraints for relational integrity
- Composite uniqueness to prevent duplicate student-job applications
- NOT NULL constraints applied to mandatory fields to maintain data integrity.

Relationships

- One-to-Many → Company → Job
- One-to-Many → Student → Application
- One-to-Many → Job → Application



6. Key Route Endpoints

Endpoint	Method	Purpose
/admin/login	GET/POST	Admin authentication
/company/register	GET/POST	Company registration
/company/job/create	GET/POST	Create job posting
/student/apply/<job_id>	GET	Apply for job
/admin/company/<id>/approve	GET	Approve company
/logout	GET	End user session

(Additional routes implemented for role-based workflows and administrative controls.)

7. Architecture and Features

Architecture Overview

The project follows a structured modular design:

- **app.py** — Main Flask application and route controllers
- **models.py** — Database models using SQLAlchemy ORM
- **templates/** — Jinja2 templates for HTML pages and Bootstrap 5 for responsive UI and layout design
- **database/** — SQLite database storage

Core Features

- Role-based authentication (Admin / Company / Student)
- Session-based authentication using Flask session handling
- Company registration and admin approval workflow
- Job posting and approval pipeline
- Student job browsing and application system
- Application status tracking
- Student profile management
- Admin dashboards for monitoring platform activity

Additional Features

- Unique application validation
 - Dashboard analytics and summaries
 - Clean UI and UX
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8. Video Presentation

Drive Link:  MAD_1_Presentation.mp4

End of Report